

Milestone 5 – Childcare Prices Analysis and Affordability in the United States

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Project Overview

This project involves analysis of the National Database of Childcare-Prices to comprehend how the costs of childcare differ in United States & how these costs differ depending on geographic and economic differences. It has evolved from an exploratory data analysis to creating three mediums of communication: an interactive dashboard, a static infographic, and a narrated presentation.

The analysis was conducted by first exploring data using Python and created visualizations which shows patterns in the costs of childcare within states & counties. This final version has been developed based on feedback provided by the instructor throughout the milestones of the project and has improved layouts, visuals, and storytelling to communicate the insights more clearly.

Summary of Analysis

Python was used to for data cleaning, analysis, & creating visualizations. The analysis was conducted on center-based childcare prices for infants, toddlers, pre-schoolers at both county and state levels. The major processes that were carried out during the data analysis include selectively dealing with missing values, computing statistical measures, and exploring correlations between childcare prices and median household income as of 2018.

Several data visualizations is developed during the course of the analysis. These include bar plots that show the states with the highest infant childcare prices, histograms that show the distribution of childcare prices among counties, and scatter plots that show the comparison between different levels of household income and the costs of childcare.

The major aim of the data analysis was to create descriptive insights as objected to create predictive models. The aim was to communicate the patterns of childcare costs to policymakers and non technical individuals.

Key Findings

From this analysis, it is clear that there is a considerable difference in childcare costs in different regions of the United States. Infant care is found to be the highest in compare to other care in any region. Regions - District of Columbia, Massachusetts, Hawaii, and California are found to have high costs in infant childcare.

From the distribution of childcare costs in different counties, it is clear that some regions have moderate levels of childcare costs while others have high costs in comparison to others.

From this analysis, it is clear that median household income has a moderate positive relationship with childcare costs in different regions of the United States. It is clear that regions with high median household income levels also have high childcare costs. However, it is also clear that even in regions with high income levels, childcare costs are a considerable burden.

From this analysis, it is clear that childcare costs vary significantly in different regions of the US, depending on regional economic factors and geographical factors.

Mediums

Three communication mediums were developed as below.

Interactive Dashboard (Tableau)

To provide interactive exploration of the patterns, an interactive dashboard has been created using Tableau. Various visualizations, such as bar charts representing the top states with the highest price for infant childcare, a histogram representing the distribution of the price for infant care, and a scatter plot representing the price for childcare relative to income for each family, are included. This medium is for policymakers, researchers, and analysts, as they benefit from interactive exploration of the data.

Note: Dashboard is submitted as a separate file.

Static Infographic (Canva)

A static infographic was designed to present the most important findings in a single visual format. Charts produced using Python were integrated into the infographic using Canva for better readability and visualization. The infographic presents important information such as the variation of costs at the state level, the distribution of childcare costs, and the association between income and childcare costs. This format could be good for the general public, parents, organizations that may need a summary of childcare costs.

Note: Canva is submitted as a separate file.

Narrated Presentation (PowerPoint)

The presentation follows a structured format in explaining the analysis through a brief presentation of a series of slides. It includes visualizations used in the dashboard and infographic. It discusses the data set, key findings, and implications of the differences in childcare costs. This medium could be best for stakeholders who are best served by a guided explanation of data.

Note: PPT with narration is submitted as a separate file.

Design Decisions

The design choices were centered on clarity, simplicity, as ease of access for all media. A color scheme has been maintained throughout the design to ensure visual consistency. Simple forms of charts such as bar charts, histograms, scatter charts were preferred for the ease of interpretation of the information. Considerable thought has been put into the spacing, labeling, and presentation of the visualizations to ensure ease of interpretation for the audience. Consistency in the use of fonts, colors is maintained across all media.

Ethical Considerations

The dataset used in the project is publicly available and does not include any identifiable information. This minimizes any potential legal issues. Data preparation included the elimination of data that included missing information in the childcare prices. No changes were made to the data other than aggregation.

One of the risks of presenting aggregated data is the oversimplification of the problem. For example, the averages for the states could hide the actual data for the counties. Additionally, the relationship between income level and childcare prices is not causal. These issues were mitigated by the use of proper labeling of the visualization

Lessons Learned

This project also showed the significance of matching the level of analysis with the audience to be addressed. With this project, the early stages were focused on the analysis of the data, while the later stages were focused on the storytelling. This feedback from the instructor was helpful in making the visuals more understandable while ensuring that the mediums used were effective in communicating the ideas obtained from the analysis of the data. If the project had to be taken to the next notch, the affordability factor could be analyzed based on the cost of childcare as a function of income.

Conclusion

This project demonstrates the potential for data visualization to effectively communicate complex economic topics such as the issue of childcare costs. By creating interactive dashboard, infographic canva, and a presentation, the project aims to provide an effective overview of the price variation of childcare costs throughout the United

States. By utilizing these mediums, the project demonstrates an effective approach for communicating the results while emphasizing the importance of data visualization ethics.